

Dietary manganese intake and type of lipid do not affect clinical or neuropsychological measures in healthy young women.

Because manganese (Mn) is potentially toxic, and because dietary fat type may affect Mn absorption, the objectives of the current study were to determine whether diets containing very low or very high amounts of Mn and enriched in either saturated or unsaturated fats affected measures of neuropsychological and basic metabolic function. Healthy young women were fed for 8 wk each, in a crossover design, diets that provided 0.8 or 20 mg of Mn/d. One half of the subjects received 15% of energy as cocoa butter, and one half received 15% of energy as corn oil. A meal containing (54)Mn was fed after 4 wk, and subjects underwent whole-body counting for the next 21 d. Blood draws and neuropsychological tests were administered at regular intervals during the dietary periods. When subjects consumed the diets low in Mn, compared with the high Mn diets, they absorbed a significantly higher percentage of (54)Mn, but had a significantly longer biological half-life of the absorbed (54)Mn. Manganese intake did not affect any neurological measures and only minimally affected psychologic variables. These data show that efficient mechanisms operate to maintain Mn homeostasis over the range of intakes that may be encountered in a mixed Western diet. Thus, dietary intakes of Mn from 0.8 to 20 mg for 8 wk likely do not result in Mn deficiency or toxicity signs in healthy adults.